

EXPERIMENTS 6 – 10

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EXPERIMENT 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

CODE:

```
# Dataset
```

```
product <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
```

```
# 1. Grouped Bar Chart
```

```
sales <- as.matrix(product[,2:4])
barplot(sales,
  beside=TRUE,
  col=c("red","blue","green"),
```

```

names.arg=product$Product,
xlab="Products",
ylab="Sales",
main="Quarter 1 Product Sales")
legend("topright",
      legend=c("January","February","March"),
      fill=c("red","blue","green"))

```

2. Stacked Area Chart

```

months <- c(1,2,3)
matplot(months, t(sales),
        type="l",
        lwd=2,
        col=c("red","blue","green"),
        xaxt="n",
        xlab="Months",
        ylab="Sales",
        main="Sales Trend")
axis(1, at=1:3,
     labels=c("January","February","March"))
legend("topleft",
      legend=product$Product,
      col=c("red","blue","green"),
      lty=1)

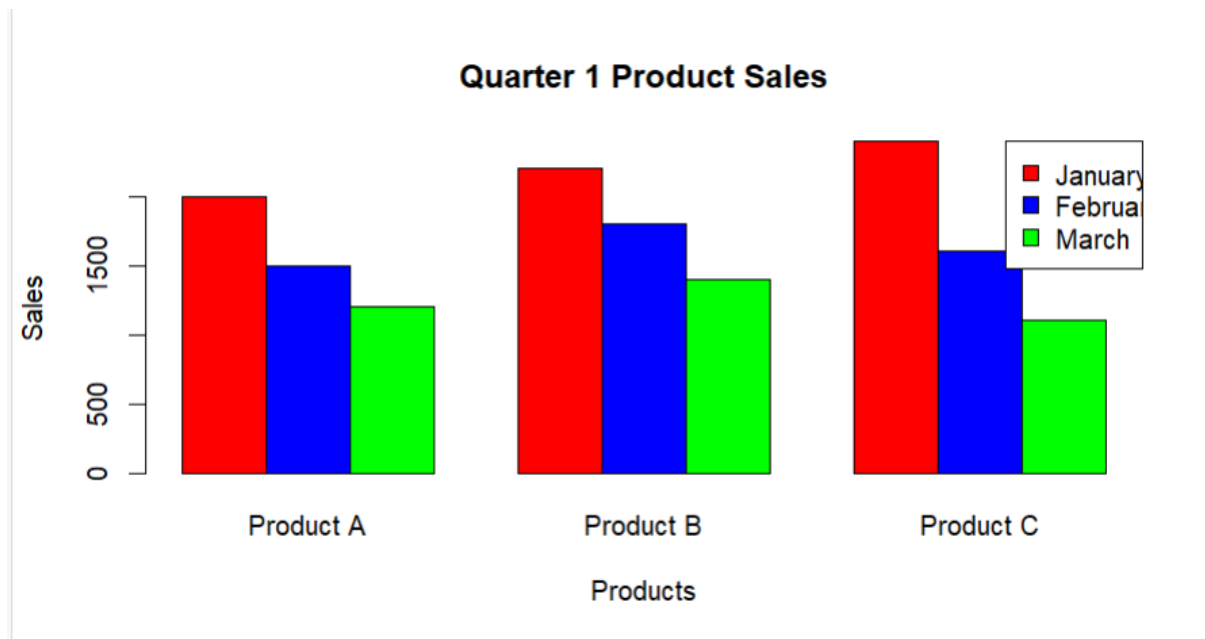
```

3. Table

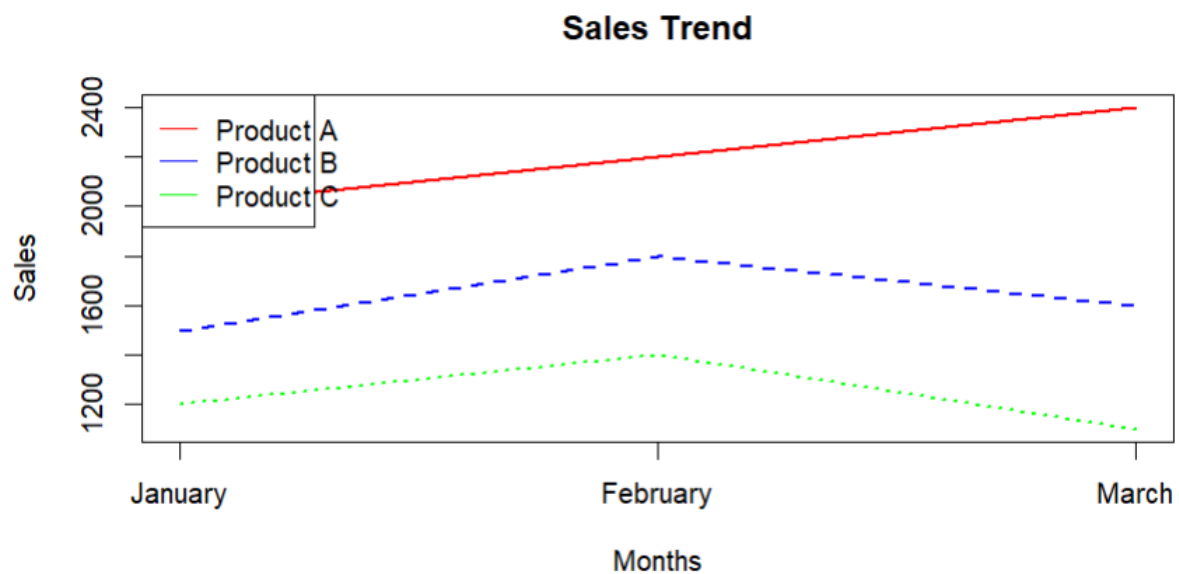
Product

QUESTIONS:

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.



3. Build a table to show the monthly sales data for each product. Label the table elements.

```
#-----
# 3. Table
#-----
product
  Product January February March
Product A   2000    2200   2400
Product B   1500    1800   1600
Product C   1200    1400   1100
```

EXPERIMENT 7

Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

CODE:

```
# Customer Demographics
```

```
customer <- data.frame(
```

```
  CustomerID=c(1,2,3),
```

```
  Age=c(28,35,42),
```

```
  Gender=c("Female","Male","Female"),
```

```
  Income=c(50000,60000,75000)
```

```
)
```

```
# 1. Age Distribution Bar Chart
```

```
barplot(customer$Age,
```

```
  names.arg=customer$CustomerID,
```

```
  col="skyblue",
```

```
  xlab="Customer ID",
```

```
  ylab="Age",
```

```
  main="Customer Age Distribution")
```

```
# 2. Gender Pie Chart
```

```
gender_count <- table(customer$Gender)
```

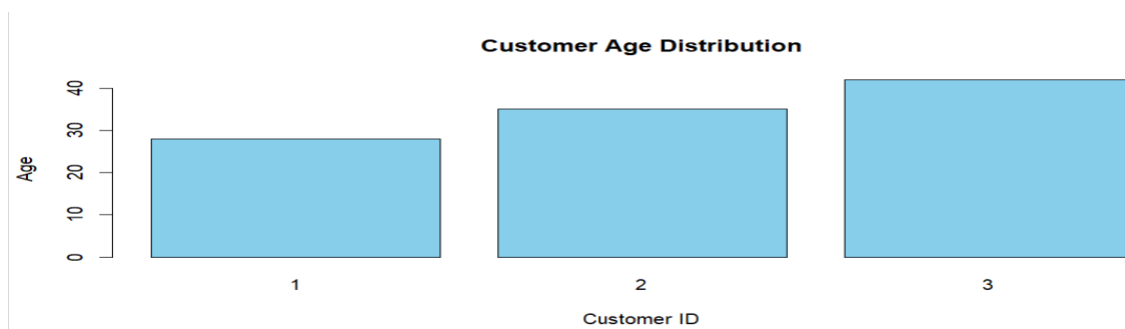
```
pie(gender_count,
    col=c("pink","lightblue"),
    main="Gender Distribution")
```

3. Table

Customer

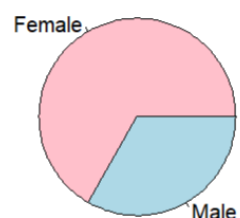
QUESTIONS

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.



2. Generate a pie chart to represent the distribution of customers by gender.

Gender Distribution



3. Build a table to show the demographic information for each customer. Label the table elements.

```
#-----
# 3. Table
#-----
customer
CustomerID Age Gender Income
1 28 Female 50000
2 35 Male 60000
3 42 Female 75000
```

EXPERIMENT 8

Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

CODE:

```
# Employee Performance
```

```
employee <- data.frame(
```

```
  EmployeeID=c(1,2,3),
```

```
  Department=c("Sales","HR","Marketing"),
```

```
  Years=c(5,3,7),
```

```
  Performance=c(85,92,78)
```

```
)
```

```
# 1. Line Chart
```

```
plot(employee$Years,
```

```
  employee$Performance,
```

```
  type="o",
```

```
  pch=16,
```

```
  col="blue",
```

```
  xlab="Years of Service",
```

```
  ylab="Performance Score",
```

```
  main="Employee Performance Trend")
```

```
# 2. Department Bar Chart
```

```
dept <- table(employee$Department)

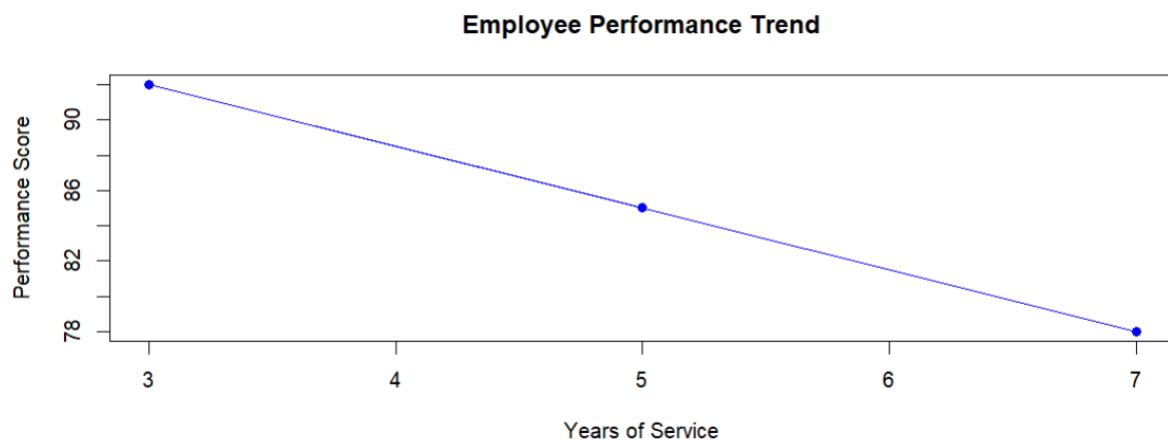
barplot(dept,
        col="orange",
        xlab="Department",
        ylab="Number of Employees",
        main="Employees by Department")
```

3. Table

Employee

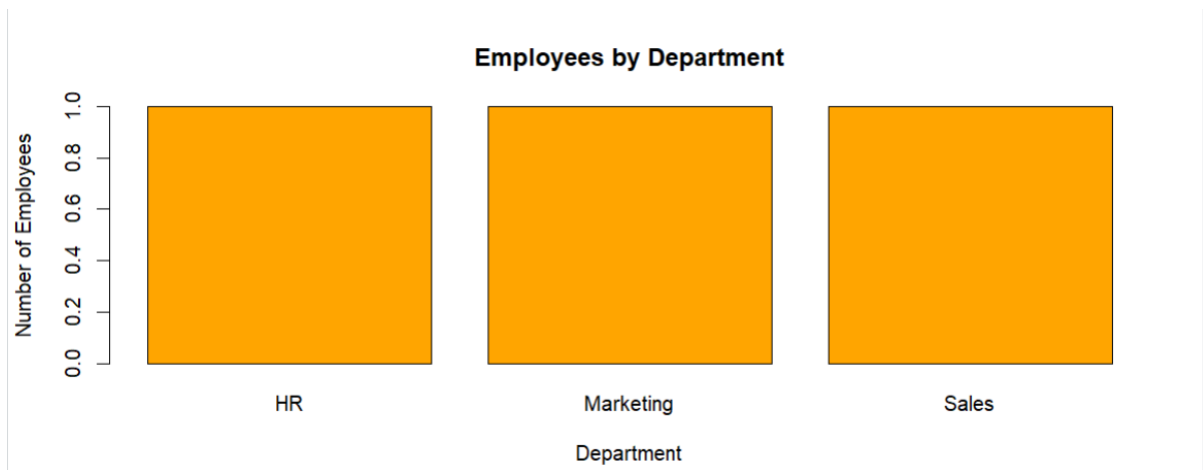
QUESTIONS:

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.



2. Generate a bar chart showing the distribution of employees across different departments.

Label the chart elements.



3. Build a table to display the performance data for each employee. Label the table elements.

```
#-----  
# 3. Table  
#-----  
employee  
EmployeeID Department Years Performance  
1 Sales 5 85  
2 HR 3 92  
3 Marketing 7 78
```

EXPERIMENT 9

Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in \$)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

CODE

```
# Product Inventory
```

```
inventory <- data.frame(
  Product=c("Product A","Product B","Product C"),
  Quantity=c(250,175,300),
  Price=c(20,15,18)
)
```

```
# 1. Quantity Bar Chart
```

```
barplot(inventory$Quantity,
  names.arg=inventory$Product,
  col="green",
  xlab="Products",
  ylab="Quantity",
  main="Inventory Quantity")
```

```
# 2. Stacked Bar Chart
```

```
data <- rbind(inventory$Quantity,
  inventory$Price)
barplot(data,
```

```

col=c("blue","red"),
names.arg=inventory$Product,
xlab="Products",
ylab="Values",
main="Quantity and Price")
legend("topright",
      legend=c("Quantity","Price"),
      fill=c("blue","red"))

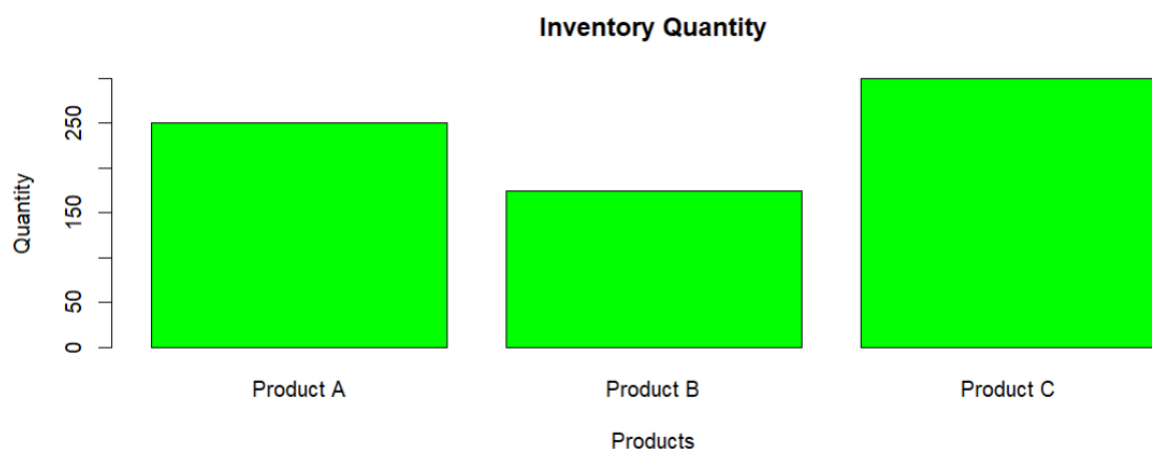
```

3. Table

Inventory

QUESTIONS:

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.



2. Generate a stacked bar chart to show the quantity of each product within different product categories.



3. Build a table to show the inventory data for each product. Label the table elements.

```
#-----
# 3. Table
#-----
inventory
  Product Quantity Price
Product A      250    20
Product B      175    15
Product C      300    18
```

EXPERIMENT 10

Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

CODE:

```
# Survey Responses
survey <- data.frame(
  SurveyID=c(1,2,3),
  Q1=c("A","B","C"),
```

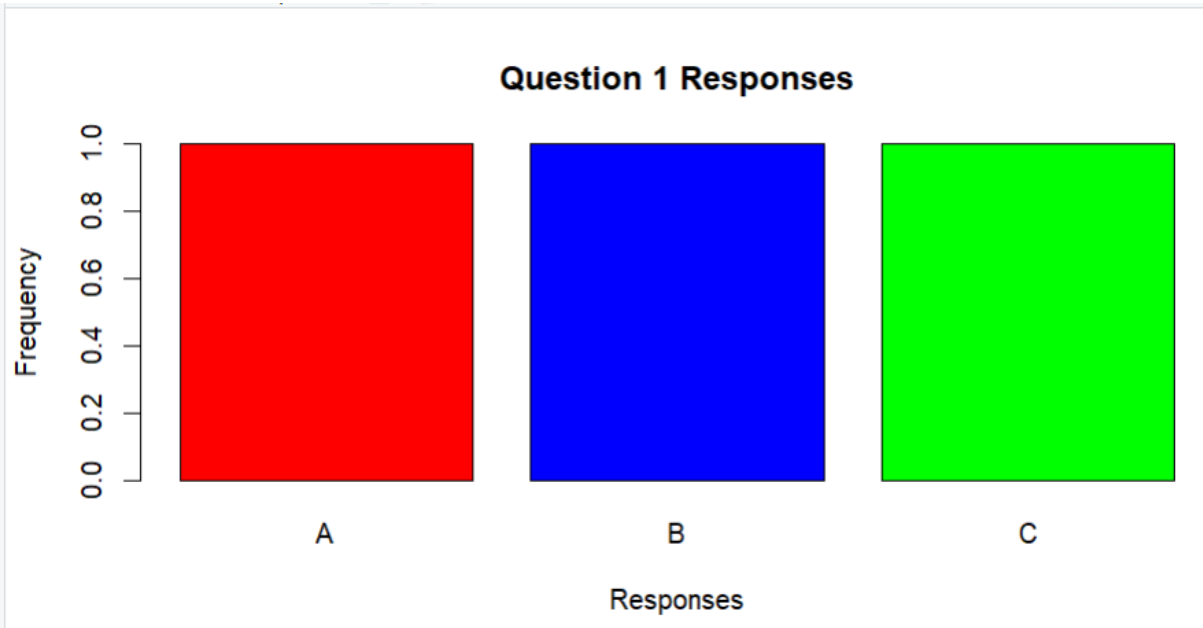
```

Q2=c("B","A","A"),
Q3=c("C","D","B")
)
# 1. Grouped Bar Chart (Q1)
q1 <- table(survey$Q1)
barplot(q1,
        col=c("red","blue","green"),
        xlab="Responses",
        ylab="Frequency",
        main="Question 1 Responses")
# 2. Stacked Bar Chart
responses <- rbind(
  table(factor(survey$Q1, levels=c("A","B","C","D"))),
  table(factor(survey$Q2, levels=c("A","B","C","D"))),
  table(factor(survey$Q3, levels=c("A","B","C","D")))
)
barplot(t(responses),
        beside=FALSE,
        col=c("red","blue","green"),
        xlab="Responses",
        ylab="Count",
        main="Survey Responses")
legend("topright",
       legend=c("Question1","Question2","Question3"),
       fill=c("red","blue","green"))
# 3. Table
survey

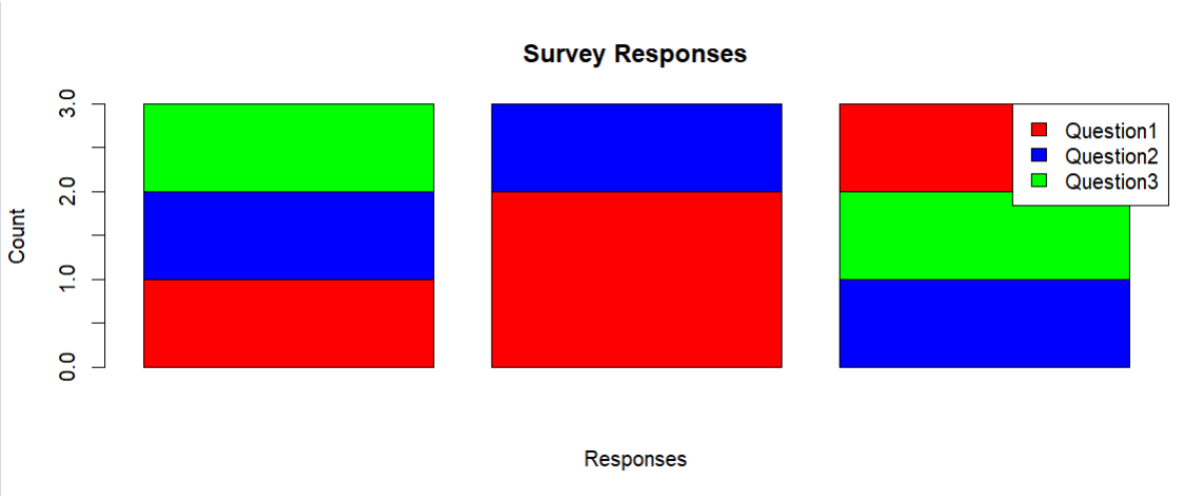
```

QUESTIONS:

1. Create a grouped bar chart to visualize the distribution of answers for Question 1.
Label the chart elements.



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.



3. Build a table to show the survey response data for each survey. Label the table elements.

```
#-----  
# 3. Table  
#-----  
survey  
SurveyID Q1 Q2 Q3  
1 A B C  
2 B A D  
3 C A B
```